

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Conservation priorities, legislation & protocols - Set 1

Mark scheme

AQA spec 3.1.2.3.1-3.1.2.3.2; The living environment; textbook pp. 26-39

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Coordinating global data on biodiversity conservation (Red List) (1) • Increasing understanding of the importance of biodiversity (1) • Deploying nature-based solutions to global challenges in climate, food and sustainable development (1)	3
01.2	<i>1 mark for each point.</i> • Decline = $(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%$ (1) • Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km²) and adults (2100) only meet Endangered (1) • A species is placed in the highest category for which it meets any one criterion, so Critically Endangered (1)	3
01.3	<i>1 mark for each point.</i> • Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed (1) • Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests) (1) • Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations (1)	3
01.4	<i>1 mark.</i> • Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.	

Question	Marking guidance	Marks
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • EDGE: Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis); loss would remove a unique branch of evolution / genes (2) • Endemic: found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur); if lost locally there is no population elsewhere to recolonise (2) • Keystone: disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers); losing them causes loss of many other species (2) • Flagship: high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths) (2)	8
02.2	<i>1 mark for each point.</i> • A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large (1) • A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.	
03.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Designation of protected areas (SSSIs, MCZs) (1) • Protection of wild birds and their nests (except some pests / game birds) (1) • Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts (1) • Uprooting wild plants generally illegal (1) • Bats must not be disturbed; roof timber treatments must be non-toxic to bats (1)	4
03.2	<i>1 mark for each point.</i> • Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding (1) • Examples: great apes, big cats, rhinos, blue whale (1) • Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected (1) • Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap (1)	4
03.3	<i>1 mark for each point.</i> • Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market (1) • Removing the profit removes the incentive to collect (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) (1) • SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) (1) • Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally (1) • MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities (1)	4
04.2	<i>1 mark for each point.</i> • Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) (1) • Management agreements between owners and statutory bodies to maintain suitable conditions (1) • But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation (1)	3
04.3	<i>1 mark for each point.</i> • Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues (1) • Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) (1) • Benefits everyone so society pays through central funding; legislation cannot compel positive management (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
05.1	<i>1 mark for the correct answer.</i> • $(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$ (1)	1
05.2	<i>1 mark for each point.</i> • Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly (1) • Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades (1) • No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found (1)	3

Question	Marking guidance	Marks
05.3	<i>1 mark for each point.</i> • Moratorium on commercial whaling since 1986 until populations recover (1) • Total protection for certain species; whale sanctuaries (1) • Limits on numbers and sizes taken; protection of suckling mothers and calves (1) • Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas (1)	4
05.4	<i>1 mark for each point.</i> • ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since (1) • Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	